

No usable cusp in the magnetic stress correlator: the turbulent gravitational-wave peak is a window effect

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The gravitational-wave spectrum radiated by primordial turbulence is fixed, at fixed spatial source, by the unequal-time correlator (UETC) of the anisotropic stress. Whether that correlator is differentiable at zero time lag decides the shape of the spectrum above its peak: a cusp gives a heavy ω^{-2} tail and displaces the peak to the source scale, while a smooth correlator does neither. The velocity UETC of hydrodynamic turbulence has been measured and is Gaussian, hence cusplike; the corresponding magnetic *stress* correlator, which is what actually sources the waves, has not been. We show that it can be constrained without any two-time data. In a magnetohydrodynamic simulation started from rest, the gravitational-wave field is itself a running cosine transform of the stress UETC evaluated on the light cone, so the radiated energy at the end of the run is a parameter-free *prediction* for each candidate correlator once the stress and the Alfvén speed are read off. Applied to a published run, cusplike correlators reproduce the radiated energy while cusped ones overpredict it by one to two orders of magnitude. The magnetic stress correlator is therefore no more cusped than the hydrodynamic one, the ω^{-2} tail is excluded for this class of source, and the source-scale peak seen in simulations is produced by the finite emission window rather than by decorrelation. The method needs only quantities such simulations already write to disk.

I.	INTRODUCTION
II.	THE CRITERION
III.	THE MEASUREMENT
IV.	CONSEQUENCES
V.	SUMMARY

correlator	$E_{\text{model}}/E_{\text{data}}$
coherent, $f = 1$	0.35–1.19
Gaussian sweeping, $f = e^{-(kv_A\tau)^2/2}$	0.90–1.14
cusped, $f = e^{-kv_A \tau }$	19.5–97.8
BK2016, $f = (1 + kv_A\tau)^{-2/3}$	13.0–64.0

TABLE I. Parameter-free forward models of the radiated energy over $p = 1.5$ –30.

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